

[Home](#) > [My Courses](#) > [Financial Markets](#) > [M3: Correlation](#) > FM Graded Quiz M3

FM Graded Quiz M3

Question 1

Calculate the Sharpe ratio for a portfolio with return 8%, risk-free rate 1%, and standard deviation 5.162%.

- ☐ 1.215
- ☐ 1.356
- ☐ 1.3
- ☐ 1.4



Time left:
58:29

QUESTIONS

X	X	X	X
X	X	X	X
X	X	X	X
X	X	X	X

Question 2

What is the primary difference between mark-to-market and mark-to-model pricing?

- ☐ Mark-to-market is used for bonds, while mark-to-model is used for stocks
- ☐ Mark-to-market uses observable market prices, while mark-to-model uses estimated prices
- ☐ Mark-to-market is always more accurate than mark-to-model
- ☐ Mark-to-market is used for long positions, while mark-to-model is used for short positions

Question 3

Which correlation measure is most appropriate for assessing the agreement between two stock analysts' rankings?

- ☐ Spearman correlation
- ☐ Pearson correlation
- ☐ Linear correlation
- ☐ Kendall's tau

Question 4

If two assets have a correlation of 0.5 and individual volatilities of 15% and 20% what is their covariance?

- ☐ 0.015
- ☐ 0.02
- ☐ 0.01
- ☐ 0.025

Question 5

What is the potential impact of a stock being removed from a major index on its ETF ownership?

- ☐ It has no effect on the stock's ownership or price
- ☐ It only affects the stock's short-term trading volume
- ☐ It always leads to increased demand for the removed stock
- ☐ It may lead to decreased demand and potentially lower prices for the removed stock

Question 6

Assume that a portfolio consists of two equally weighted assets having returns of 5% and 11%, standard deviations of 3% and 8% respectively,

and the correlation between the two assets is 1. What is the reduction in volatility if the correlation was -0.5 instead?

- ☐ -0.02
- ☐ 0.005
- ☐ 0.01
- ☐ 0.02

Question 7

What statistical technique exploits high correlations in a correlation matrix?

- ☐ Cluster analysis
- ☐ Time series analysis
- ☐ Multiple regression analysis
- ☐ Principal component analysis

Question 8

An ETF has a beta of 1.2 relative to its benchmark index. If the index returns 10% in a year, what is the expected return of the ETF before expenses?

- ☐ 14.40%
- ☐ 10%
- ☐ 8.33%
- ☐ 12%

Question 9

Calculate the portfolio standard deviation given:

$$w_A = 0.5, w_B = 0.5, \sigma_A = 3\%, \sigma_B = 8\%, \rho_{AB} = 0.7$$

- ☐ 5.00%
- ☐ 4.89%

- ☐ 5.50%
- ☐ 5.16%

Question 10

What is the Coefficient of Variation (CV) used for in portfolio analysis?

- ☐ To determine the optimal asset allocation
- ☐ To calculate the risk-free rate
- ☐ To compare volatility per unit of return across different portfolios
- ☐ To measure the correlation between assets

Question 11

Which of the following statements about correlation is false?

- ☐ Correlation always implies causation
- ☐ A correlation of 0 indicates no linear relationship
- ☐ Perfect negative correlation results in the lowest possible portfolio variance
- ☐ Correlation can range from -1 to 1

Question 12

If the NASDAQ declined 3% in a day and a \$100000 portfolio tracked it perfectly what would be the new portfolio value?

- ☐ \$96,500
- ☐ \$97,500
- ☐ \$97,000
- ☐ \$98,000

Question 13

What strategy could an investor use to potentially benefit from the differences between active and passive ETF management?

- ☐ Avoid ETFs altogether and invest only in individual stocks
- ☐ Combine low-cost passive ETFs for core holdings with select active ETFs for potential outperformance in specific sectors
- ☐ Invest only in actively managed ETFs for all sectors
- ☐ Always choose the ETF with the lowest expense ratio

Question 14

According to "OpenIntro Statistics", if the correlation between travel time in miles and distance in minutes is $r = 0.636$ what would be the correlation if travel time was measured in kilometers and distance in hours?

- ☐ 0.318
- ☐ 0.636
- ☐ 0
- ☐ 1.272

Question 15

Design a hypothetical asset allocation ETF that could potentially outperform in both high and low volatility environments. What would be its key features?

- ☐ Dynamic allocation based on volatility levels with a mix of uncorrelated assets
- ☐ 100% investment in a single sector
- ☐ Exclusively short-term government bonds
- ☐ Fixed 50-50 split between stocks and bonds

Question 16

If a US-exchange-traded ETF's NAV is calculated at 11:30 AM ET due to tracking a European index and the US exchange closes at 4:00 PM ET, what time frame is considered the NAV of this ETF 'stale'?

- ☐ 11:30 AM ET - 4:00 PM ET
- ☐ 12:00 PM ET - 4:00 PM ET
- ☐ 9:30 AM ET - 11:30 AM ET
- ☐ 9:30 AM ET - 4:00 PM ET

SUBMIT